

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO3: Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)

Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:30

Annexure A

Case study

Code of Conduct:

I __P Sai Bhavyatha(192424333)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Annexure A

Case study

Case Study 1: Defect Detection in Industrial Parts

A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images of the components are binary and may contain small protrusions or missing parts. **Questions:**

- How can region-based representation and medial axis computation help in identifying defective parts?
- Discuss how noise in image acquisition could affect shape analysis and propose preprocessing steps to improve accuracy.
- Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal vs. defective components.

Case Study 2: Medical Image Segmentation

A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients. **Questions:**

- Explain how active contours (snakes) can be applied to segment the organ boundaries.
- Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.
- How could level set methods improve segmentation for cases where organs split or merge in the images?

IMPLEMENTATION

CASE STUDY 1: DEFECT DETECTION IN INDUSTRIAL PARTS

Case Understanding

The manufacturing company wants to automatically identify defective gears and machine components using computer vision. The input images are binary, meaning that the object and background are represented using different pixel values. Defects such as **small protrusions, missing parts, and irregular regions** can change the shape and structure of the component.

For reliable defect detection, the system should extract meaningful shape features, reduce acquisition noise, compare the extracted features with normal components, and classify the components as **normal or defective**.

A. How can region-based representation and medial axis computation help in identifying defective parts?

Region-Based Representation

Region-based representation analyzes the complete area occupied by an object rather than considering only its boundary. For an industrial component, the binary image can be analyzed to obtain properties such as:

- Area
- Centroid
- Perimeter
- Bounding box
- Connected components
- Number and size of holes
- Region shape
- Pixel distribution

A normal gear or machine component generally has a consistent structure. When a defect occurs, properties such as area, perimeter, or the number of connected regions may change.

For example, if a portion of a gear is missing, the **area decreases** and the boundary structure changes. If an unwanted protrusion occurs, the **area and perimeter increase**.

Medial Axis Representation

The **medial axis**, also called the skeleton, provides a thin representation of the object while preserving its main structural and topological properties.

For a gear, the medial axis can represent:

- Central structure
- Teeth arrangement
- Branches
- Connectivity
- Internal holes
- Structural symmetry

A defective component can be identified by comparing its medial axis with that of a normal component.

For example:

Normal component:

Binary Image → Region Analysis → Medial Axis → Expected Structure

Defective component:

Binary Image → Region Analysis → Medial Axis → Structural Difference

If a gear tooth is missing, the corresponding change can appear in the skeleton. Similarly, an unwanted protrusion can introduce an abnormal branch or change in the skeleton.

Proposed Detection Procedure

1. Acquire the binary image of the component.
2. Remove image noise.
3. Perform connected-component analysis.
4. Extract region properties.
5. Compute the medial axis/skeleton.
6. Extract structural features from the skeleton.

7. Compare the features with a reference normal component.
8. Classify the component as normal or defective.

Analysis

Region-based representation provides **global information** about the component, while the medial axis provides **structural information**. Using both representations is more reliable than depending on only one feature.

Therefore, these techniques can detect both large defects, such as missing regions, and structural abnormalities, such as unwanted protrusions or changes in connectivity.

B. How can noise in image acquisition affect shape analysis and what preprocessing steps can improve accuracy?

Effect of Noise

Noise can be introduced during image acquisition because of:

- Camera sensor limitations
- Poor lighting
- Dust or contamination
- Electronic interference
- Incorrect thresholding
- Motion during image capture

Noise can create unwanted pixels or remove actual object pixels.

For example, isolated noise pixels may be interpreted as small objects or protrusions. Similarly, noise around the boundary can change the measured perimeter and distort the medial axis.

Noise can therefore affect:

- Area calculation
- Perimeter calculation
- Contour extraction

- Skeletonization
- Fourier descriptors
- Wavelet descriptors
- Defect classification

Proposed Preprocessing Pipeline

A suitable preprocessing pipeline is:

Input Image



Grayscale Conversion



Noise Reduction



Contrast Enhancement



Thresholding



Morphological Processing



Connected Component Analysis



Shape Feature Extraction



Defect Detection

1. Noise Reduction

A **Gaussian filter** can be used for Gaussian noise, while a **median filter** is effective for salt-and-pepper noise.

2. Thresholding

Thresholding separates the component from the background and produces a binary image.

3. Morphological Operations

Morphological operations such as:

- **Opening** – removes small unwanted objects/noise.
- **Closing** – fills small gaps and holes.
- **Erosion** – removes boundary pixels.
- **Dilation** – expands object regions.

can improve the binary representation.

4. Connected Component Analysis

Very small isolated components can be removed after connected-component analysis because they are likely to be noise rather than actual defects.

Analysis

Preprocessing is essential because defect detection should distinguish between **actual manufacturing defects and image-acquisition noise**. Excessive filtering, however, should be avoided because a real small defect could also be removed.

Therefore, preprocessing parameters should be selected carefully based on the expected size of the defects.

C. Compare the effectiveness of Fourier descriptors versus wavelet descriptors for classifying normal and defective components.

Fourier and wavelet descriptors both represent shape information, but they emphasize different characteristics.

Feature	Fourier Descriptors	Wavelet Descriptors
Representation	Frequency-domain contour representation	Multi-resolution representation
Main information	Global boundary shape	Local and global details
Boundary analysis	Excellent	Good
Local defects	Less effective	More effective
Scale analysis	Can be normalized	Naturally supports multiple scales
Noise sensitivity	Can be affected by high-frequency noise	Multi-scale analysis can help separate details
Computational complexity	Relatively simple	Moderate
Suitable for	Overall shape classification	Local defect and texture analysis

Fourier Descriptors

Fourier descriptors represent a closed contour using Fourier coefficients. Low-frequency components describe the general shape, while high-frequency components represent fine boundary variations.

They are highly effective when the difference between normal and defective components involves the **overall boundary shape**.

For example, if a component has a significantly different overall contour, Fourier descriptors can detect the difference efficiently.

Wavelet Descriptors

Wavelet descriptors provide a **multi-resolution representation**. They capture both coarse structural information and localized details.

This makes them useful for detecting small defects such as:

- Small cracks
- Small protrusions
- Missing teeth

- Local boundary irregularities

Final Analysis

For industrial defect classification, wavelet descriptors may be more effective when defects are **small and localized**, while Fourier descriptors are highly effective for **global shape differences**.

A strong solution would combine both:

Fourier Descriptors

+

Wavelet Descriptors

↓

Combined Feature Vector

↓

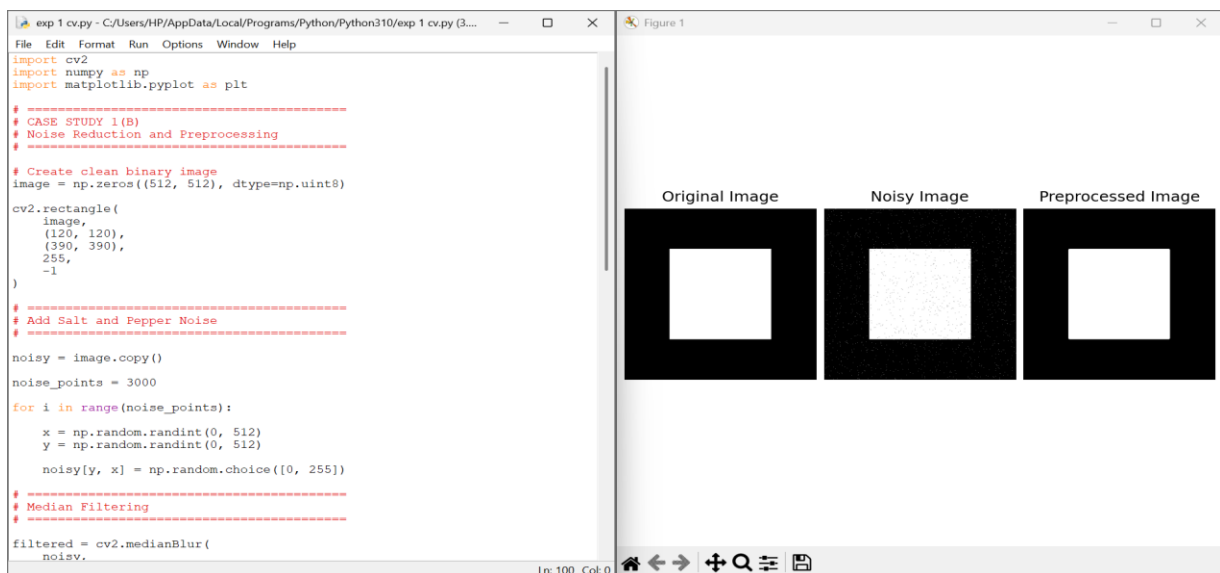
Classifier

↓

Normal / Defective

This hybrid approach can use Fourier descriptors for global shape and wavelets for local details, potentially improving classification performance.

Output:



Conclusion – Case Study 1

Region-based representation and medial axis computation provide useful structural information for detecting defects in industrial components. Proper preprocessing reduces the effect of acquisition noise. Fourier descriptors are effective for global shape differences, while wavelet descriptors are better suited to localized and multi-scale defects. Therefore, combining both types of descriptors can provide a robust industrial defect-detection system.

CASE STUDY 2: MEDICAL IMAGE SEGMENTATION

Case Understanding

The hospital wants to automatically segment organs from CT scans. The major challenge is that organs have **irregular boundaries**, and their size, shape, and position can vary between patients. Medical images can also contain noise, intensity variations, and weak boundaries.

A suitable segmentation system should therefore use deformable models that can adapt to the actual organ boundary.

A. Explain how active contours (snakes) can be applied to segment organ boundaries.

Active Contours

An **active contour**, or snake, is a deformable curve that moves toward the boundary of an object.

The contour is represented by a set of points:

The snake attempts to minimize an energy function:

where:

- controls the smoothness and continuity of the contour.
- attracts the contour toward image features such as strong edges.

Application to CT Images

The process can be performed as follows:

CT Image



Noise Reduction



Contrast Enhancement



Initial Contour



Active Contour Evolution



Organ Boundary



Segmented Organ

Step 1: Preprocessing

The CT image is first filtered to reduce noise. Contrast enhancement can be applied to make the organ boundary more visible.

Step 2: Initial Contour

An initial contour is placed around the approximate organ location.

For example, the contour can be initialized:

- Inside the organ
- Outside the organ
- Around the approximate boundary

Step 3: Contour Evolution

The snake moves iteratively toward the organ boundary. Internal forces maintain smoothness, while external image forces attract the contour toward strong boundaries.

Step 4: Final Segmentation

When the contour reaches a stable position, the enclosed region is considered the segmented organ.

Advantages

Active contours are useful because they:

- Handle irregular boundaries.
- Produce smooth segmentation.
- Incorporate image intensity and edge information.
- Can adapt to different organ shapes.
- Provide accurate boundary localization when properly initialized.

Analysis

Active contours are particularly useful when the organ boundary is not perfectly circular or rectangular. However, their success depends strongly on initialization and parameter selection.

B. Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.

1. Initialization Challenge

The initial contour must be sufficiently close to the organ.

If it is placed too far away, the snake may:

- Converge to the wrong boundary.
- Get trapped by nearby structures.
- Fail to reach the organ.
- Require many iterations.

If it is placed too close to an incorrect boundary, it may also produce inaccurate segmentation.

Possible Solution

Use automatic or semi-automatic initialization based on:

- Thresholding
 - Connected components
 - Anatomical landmarks
 - Object detection
 - Approximate organ location
-

2. Parameter Tuning

Important active-contour parameters include:

Alpha – Elasticity

Controls how strongly the contour resists stretching.

- High alpha → stronger resistance to stretching.
- Low alpha → contour can deform more easily.

Beta – Smoothness

Controls the smoothness of the contour.

- High beta → smoother contour.
- Low beta → contour can follow sharper changes.

Gamma – Step Size

Controls how quickly the contour evolves.

A very large value may cause instability, while a very small value may require many iterations.

Other Factors

Segmentation accuracy can also be affected by:

- Image noise
- Weak edges
- Low contrast
- Similar intensity between organ and surrounding tissues

- Nearby anatomical structures
- Patient-to-patient variation

Proposed Solution

A robust system can use:

1. Noise filtering.
2. Contrast enhancement.
3. Automatic initialization.
4. Appropriate parameter selection.
5. Iterative contour evolution.
6. Validation against manually segmented images.

Analysis

There is no single set of parameters that works perfectly for every patient because organ shapes and image quality vary. Therefore, adaptive parameter selection or automatic optimization can improve segmentation reliability.

C. How could level set methods improve segmentation for cases where organs split or merge?

Level Set Method

Level set methods represent a contour **implicitly** using a level set function:

The boundary is represented by:

Instead of explicitly moving individual contour points, the zero-level contour evolves according to an evolution function.

Advantage Over Traditional Active Contours

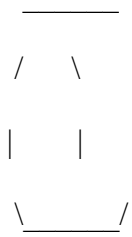
One major advantage is that level set methods can naturally handle **topological changes**.

For example:

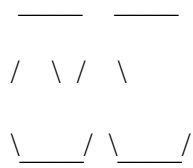
Splitting

If one segmented region needs to separate into two regions:

Before:



After:

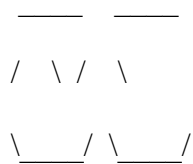


The level set representation can handle this change without manually creating a new contour.

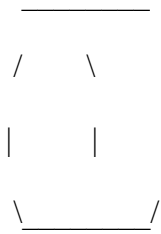
Merging

If two regions become connected:

Before



After:



The evolving level set can represent the merged boundary.

Application to Medical Images

Level set methods can be used for:

- Liver segmentation
- Heart segmentation
- Tumor segmentation
- Blood vessel analysis
- Multi-organ segmentation

Level Set Segmentation Pipeline

CT Image



Noise Reduction



Contrast Enhancement



Initial Level Set



Boundary Evolution



Topology Changes

(Splitting / Merging)



Final Segmentation

Analysis

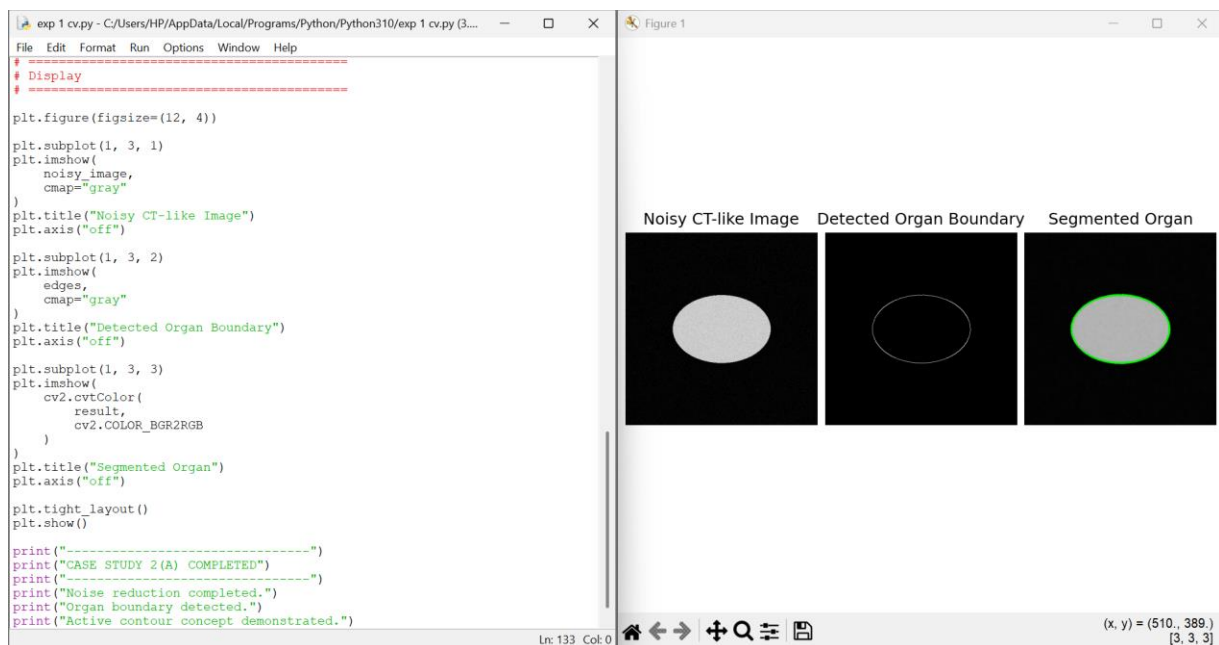
Level set methods are more flexible than traditional explicit contour representations because they can naturally handle changes in topology. This is especially important in medical images where anatomical structures can have complex shapes or appear connected/disconnected depending on the image slice.

However, level set methods can be computationally expensive and may require careful parameter selection.

Comparison of Active Contours and Level Set Methods

Feature	Active Contours	Level Set Methods
Boundary representation	Explicit curve	Implicit function
Irregular boundaries	Good	Excellent
Smooth boundaries	Good	Good
Splitting regions	Difficult	Naturally supported
Merging regions	Difficult	Naturally supported
Initialization	Important	Important
Parameter tuning	Required	Required
Computational cost	Generally lower	Can be higher
Medical segmentation	Suitable	Highly suitable for complex cases

Output:



Overall Conclusion

The two case studies demonstrate how different shape-representation and segmentation techniques can be applied to real-world computer vision problems.

For **industrial defect detection**, region-based representation provides information about the complete component, while the medial axis captures its structural characteristics.

Preprocessing is essential to prevent image noise from being incorrectly classified as defects. Fourier descriptors are effective for global shape representation, whereas wavelet descriptors provide better multi-scale and local feature information. Combining both can produce a stronger classification system.

For **medical image segmentation**, active contours provide a flexible method for locating irregular organ boundaries. However, initialization and parameter tuning are important for obtaining accurate results. Level set methods provide an additional advantage because they can naturally handle **splitting and merging boundaries**, making them suitable for complex medical segmentation tasks.

Therefore, selecting the appropriate representation or segmentation method should depend on the **type of image, nature of the object, defect characteristics, boundary complexity, noise level, and required accuracy**.

